

Cybersecurity

Haystack Dense Retrieval Practice Document

Introduction

Cybersecurity is the practice of protecting computers, networks, applications, and data from unauthorized access, misuse, disruption, modification, and destruction. Security controls help organizations reduce the likelihood and impact of cyber incidents.

CIA Triad

The CIA triad describes three important security goals: confidentiality, integrity, and availability. Confidentiality protects information from unauthorized disclosure. Integrity protects information from unauthorized modification. Availability ensures systems and data are accessible when needed.

Common Threats

Common cybersecurity threats include phishing, malware, ransomware, spyware, password attacks, social engineering, denial-of-service attacks, and data breaches. Attackers may exploit technical weaknesses or manipulate users.

Security Controls

Security controls include strong authentication, access control, encryption, firewalls, endpoint protection, secure software development, backups, patch management, logging, monitoring, and employee security awareness training.

Incident Response

A basic incident response process includes preparation, detection and analysis, containment, eradication, recovery, and lessons learned. Organizations should document incidents and improve controls based on findings.

This document is prepared as a small knowledge source for a Haystack BM25 versus dense retrieval experiment.